KAYDEN TARVARAN

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EDUCATION

University of California, San Diego Expected June 2028
B.S. in Electrical Engineering
Transfer GPA: 3.84/4.00
Relevant Coursework: Digital Design; Analog Design; Engineering Probability & Statistics; Physics I-III; C++ I-III; Calculus I-III; Circuit Analysis; Differential Equations; Linear Algebra
GitHub Portfolio: Link

SKILLS

Programming & Digital Design: SystemVerilog, RTL design, C++, Python, Git, MATLAB, NumPy, SciPy
EDA & CAD: Yosys, Icarus Verilog, LTspice, KiCad, AutoCAD, Autodesk Inventor, Fusion 360, Onshape
Hardware & Interfaces: STM32, microcontrollers, UART, SPI, I2C, digital logic, sensors, circuit prototyping, Arduino
Professional: Technical documentation, troubleshooting, teamwork, leadership, communication

PROJECTS

CNN Inference Optimization | Python, PyTorch, CUDA, NVIDIA Nsight Compute In Progress
- Benchmarking and optimizing ResNet18 on CIFAR-10 using PyTorch; analyzing CUDA kernel occupancy, GPU wave utilization, and tail effects to improve inference efficiency.

Configurable UART TX/RX Core | SystemVerilog 09/2026
- Designed synthesizable 8N1 UART TX/RX RTL in SystemVerilog with parameterized baud timing, FSM control, shift registers, and handshaking.
- Verified multiple baud rates with SystemVerilog testbenches; synthesized RTL with Yosys and debugged waveforms in Icarus Verilog and GTKWave.

FMCW Radar Signal Simulator | Python 09/2026
- Simulated 77 GHz FMCW chirps and moving targets, then estimated range, velocity, and radar cross section from complex frequency-domain data.
- Implemented peak detection and successive interference cancellation using NumPy and SciPy, accurately detecting 6 of 7 simulated targets and recovering their range, velocity, and radar cross section (RCS).

Custom Quadrotor Drone and Flight Controller PCB | KiCad 08/2026
- Designed an STM32F415 flight-controller PCB with gyroscope, barometer, and magnetometer interfaces using SPI and I2C.
- Developed battery and USB-C power architecture with automatic source selection, buck regulation, 3.3 V LDO regulation, protection, and decoupling.
- Designed and 3D printed a quadrotor frame, then integrated motors and electronic speed controllers for flight testing.

EXPERIENCE

Undergraduate Researcher | Existential Robotics Laboratory, UC San Diego 10/2026 - Present
- Investigating NaviDiffusor-based waypoint planning for autonomous race-car navigation, focusing on cost-guided diffusion for collision-aware path generation.

SPARK Academy - Electrical Design Team Member | Irvine Valley College, Irvine, CA 02/2026 - 06/2026
- Led electrical design of a regenerative-braking prototype using Arduino control logic, sensors, motor-control hardware, and iterative testing.
- Coordinated electrical/mechanical integration, documented troubleshooting, and refined the system through multidisciplinary design reviews and testing.

Engineering Intern | AGSE, Santa Fe Springs, CA 06/2024 - 08/2024
- Updated 50+ engineering drawings and 3D CAD models in AutoCAD and Autodesk Inventor, incorporating part numbers, material treatments, technical specifications, and manufacturing details for fabrication.
- Corrected 3D models and engineering documentation for ground-support equipment used with GE, Pratt & Whitney, and Rolls-Royce engine platforms.

Math Instructor | Mathnasium, Laguna Niguel, CA 03/2025 - 08/2026
- Led one-on-one and small-group K-12 instruction, identified learning gaps, and explained complex math concepts in clear steps.
- Managed several students in a fast-paced environment while communicating progress and learning needs clearly.

IT Help Desk Intern | AGSE, Santa Fe Springs, CA 06/2023 - 07/2023
- Diagnosed workstation, software, and network issues; deployed computers and documented requests, resolutions, and security procedures.